In [1]:

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Loading & viewing the data

In [2]:

```python
file_path = r'C:\Users\Neeraj Chormale\TC2 Labs\Titanic-Dataset.csv'
titanic_data = pd.read_csv(file_path)
print("First few rows of the dataset:")
print(titanic_data.head())
```

```
First few rows of the dataset:
   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

                                                Name     Sex   Age  SibSp  \
0                            Braund, Mr. Owen Harris    male  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1
2                             Heikkinen, Miss. Laina  female  26.0      0
3       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4                           Allen, Mr. William Henry    male  35.0      0

   Parch            Ticket     Fare Cabin Embarked
0      0         A/5 21171   7.2500   NaN        S
1      0          PC 17599  71.2833   C85        C
2      0  STON/O2. 3101282   7.9250   NaN        S
```

View Summary Statistics

In [3]:

```python
print("\nSummary Statistics:")
print(titanic_data.describe(include='all'))
```

```
Summary Statistics:
        PassengerId    Survived      Pclass                     Name   Sex  \
count    891.000000  891.000000  891.000000                      891   891
unique          NaN         NaN         NaN                      891     2
top             NaN         NaN         NaN  Braund, Mr. Owen Harris  male
freq            NaN         NaN         NaN                        1   577
mean     446.000000    0.383838    2.308642                      NaN   NaN
std      257.353842    0.486592    0.836071                      NaN   NaN
min        1.000000    0.000000    1.000000                      NaN   NaN
25%      223.500000    0.000000    2.000000                      NaN   NaN
50%      446.000000    0.000000    3.000000                      NaN   NaN
75%      668.500000    1.000000    3.000000                      NaN   NaN
max      891.000000    1.000000    3.000000                      NaN   NaN

               Age       SibSp       Parch  Ticket        Fare    Cabin  \
count   714.000000  891.000000  891.000000     891  891.000000      204
unique         NaN         NaN         NaN     681         NaN      147
top            NaN         NaN         NaN  347082         NaN  B96 B98
```

Check For Missing Values & it's Heatmap

In [4]:

```python
print("\nMissing Values:")
print(titanic_data.isnull().sum())
```

```
Missing Values:
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

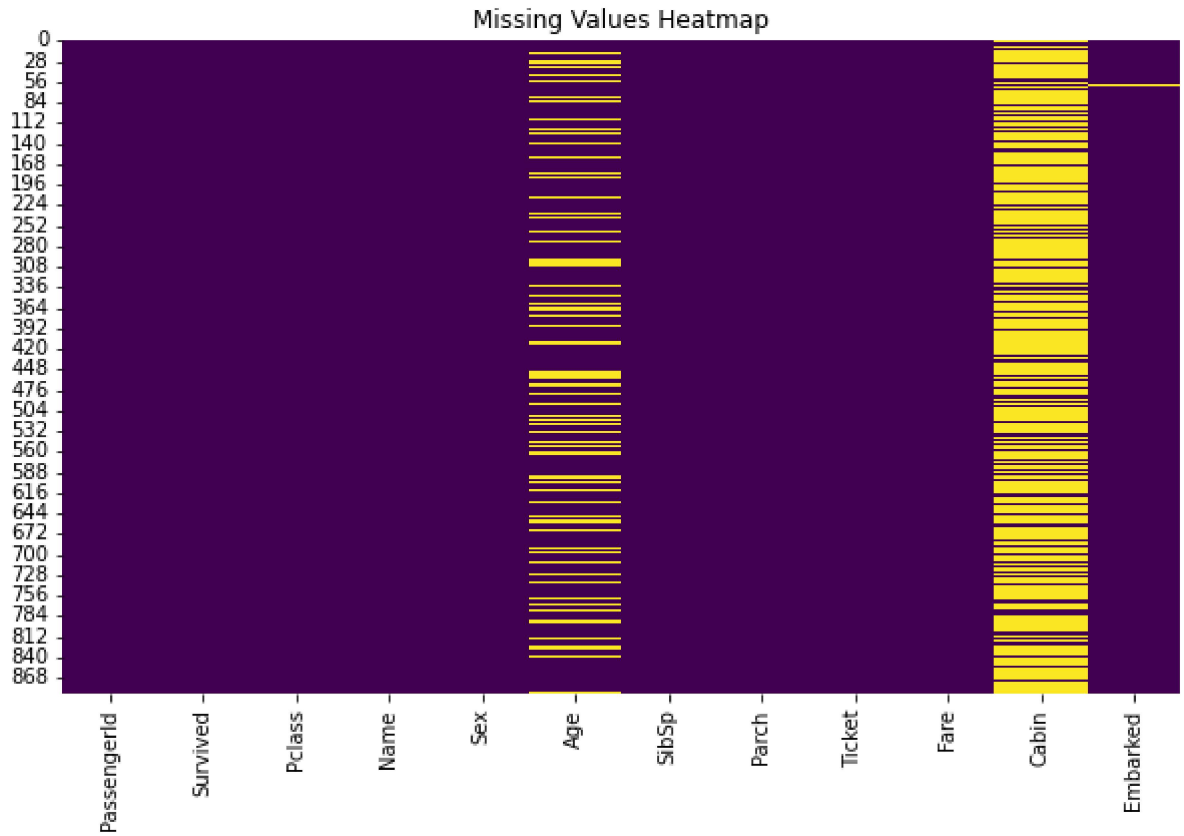
In [5]:

```
plt.figure(figsize=(10, 6))

sns.heatmap(titanic_data.isnull(), cbar=False, cmap='viridis')

plt.title('Missing Values Heatmap')

plt.show()
```



Pre-Processing Dataset

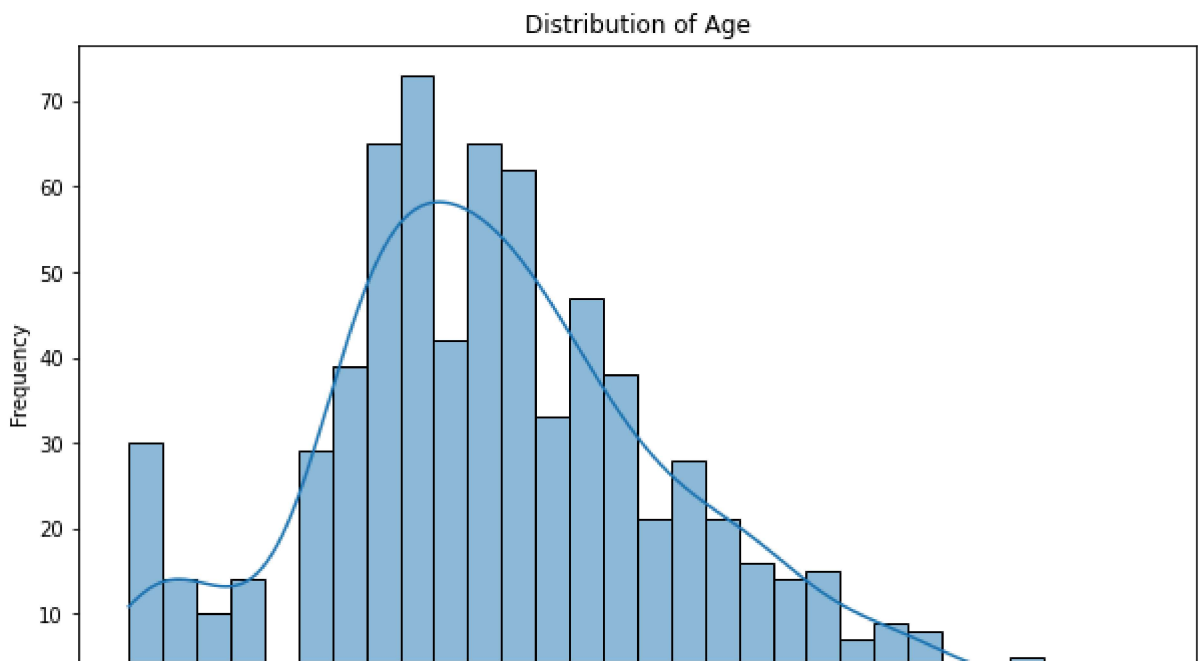
In [ ]:

```
titanic_data['Age'].fillna(titanic_data['Age'].median(), inplace=True)
titanic_data.drop(columns=['Cabin'], inplace=True)
titanic_data['Sex'] = titanic_data['Sex'].map({'male': 0, 'female': 1})
titanic_data['Embarked'] = titanic_data['Embarked'].map({'C': 0, 'Q': 1, 'S': 2})
titanic_data.drop(columns=['Name', 'Ticket', 'PassengerId'], inplace=True)
```

Age Distribution

In [6]:

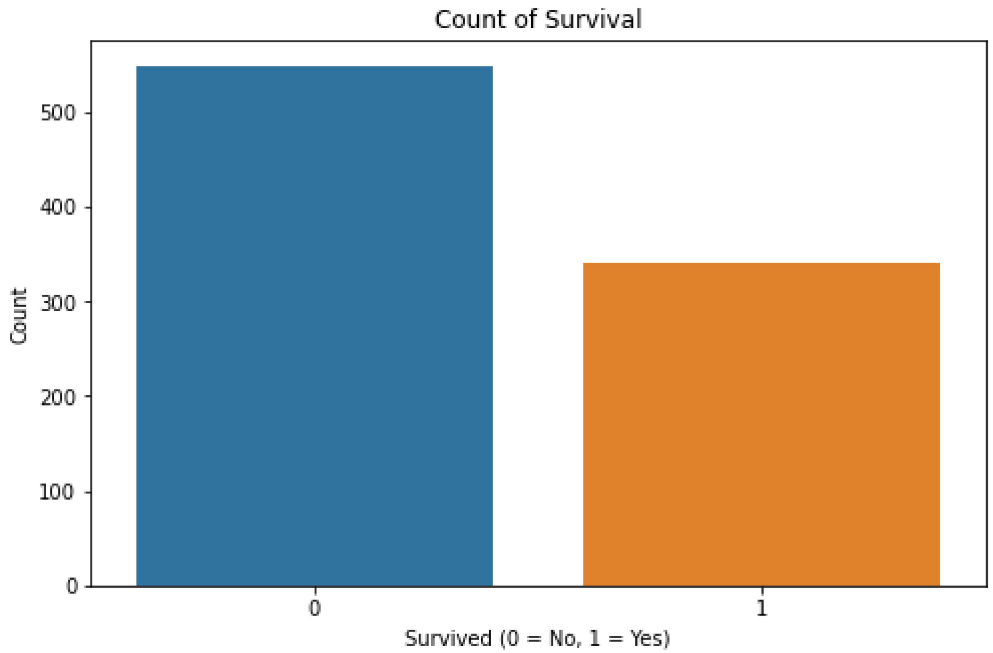
```
plt.figure(figsize=(10, 6))
sns.histplot(titanic_data['Age'], bins=30, kde=True)
plt.title('Distribution of Age')
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.show()
```



Survival Counts

In [7]:

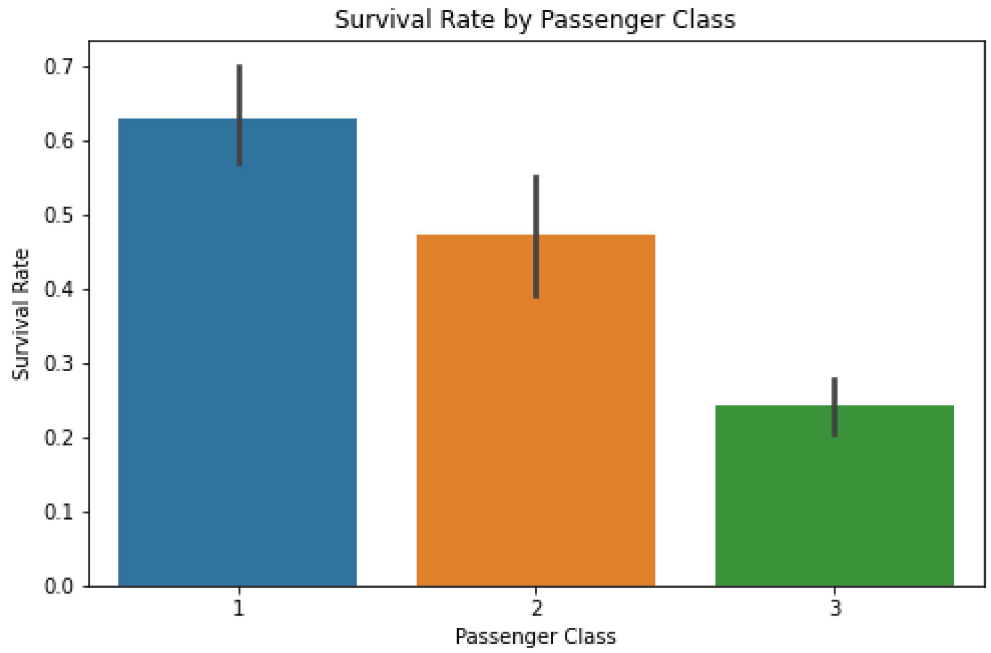
```
plt.figure(figsize=(8, 5))
sns.countplot(x='Survived', data=titanic_data)
plt.title('Count of Survival')
plt.xlabel('Survived (0 = No, 1 = Yes)')
plt.ylabel('Count')
plt.show()
```



Survival Rate by Passenger Class

In [9]:

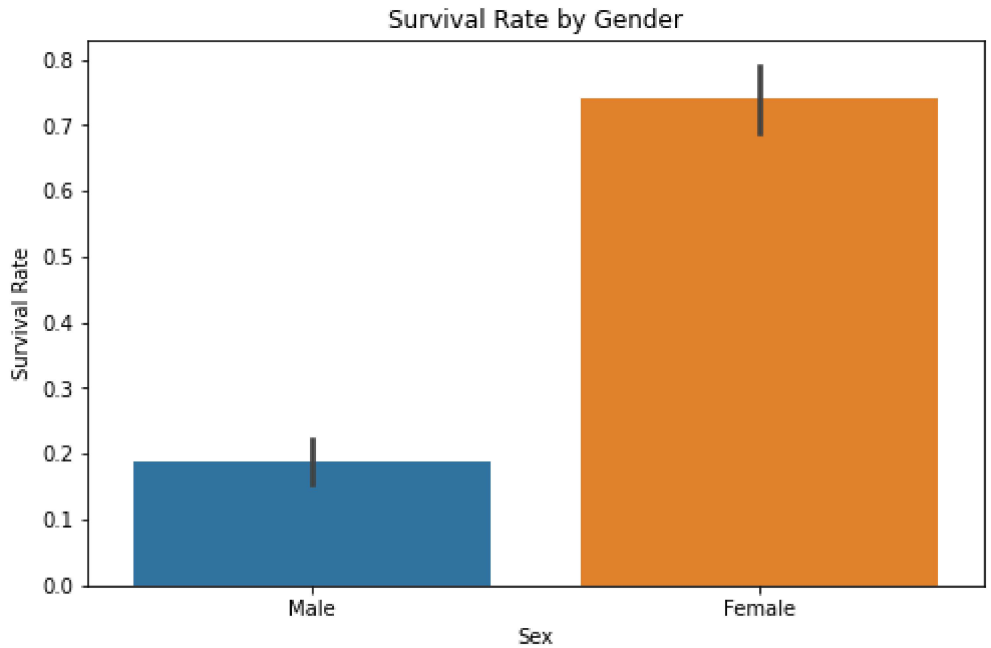
```
plt.figure(figsize=(8, 5))
sns.barplot(x='Pclass', y='Survived', data=titanic_data)
plt.title('Survival Rate by Passenger Class')
plt.xlabel('Passenger Class')
plt.ylabel('Survival Rate')
plt.show()
```



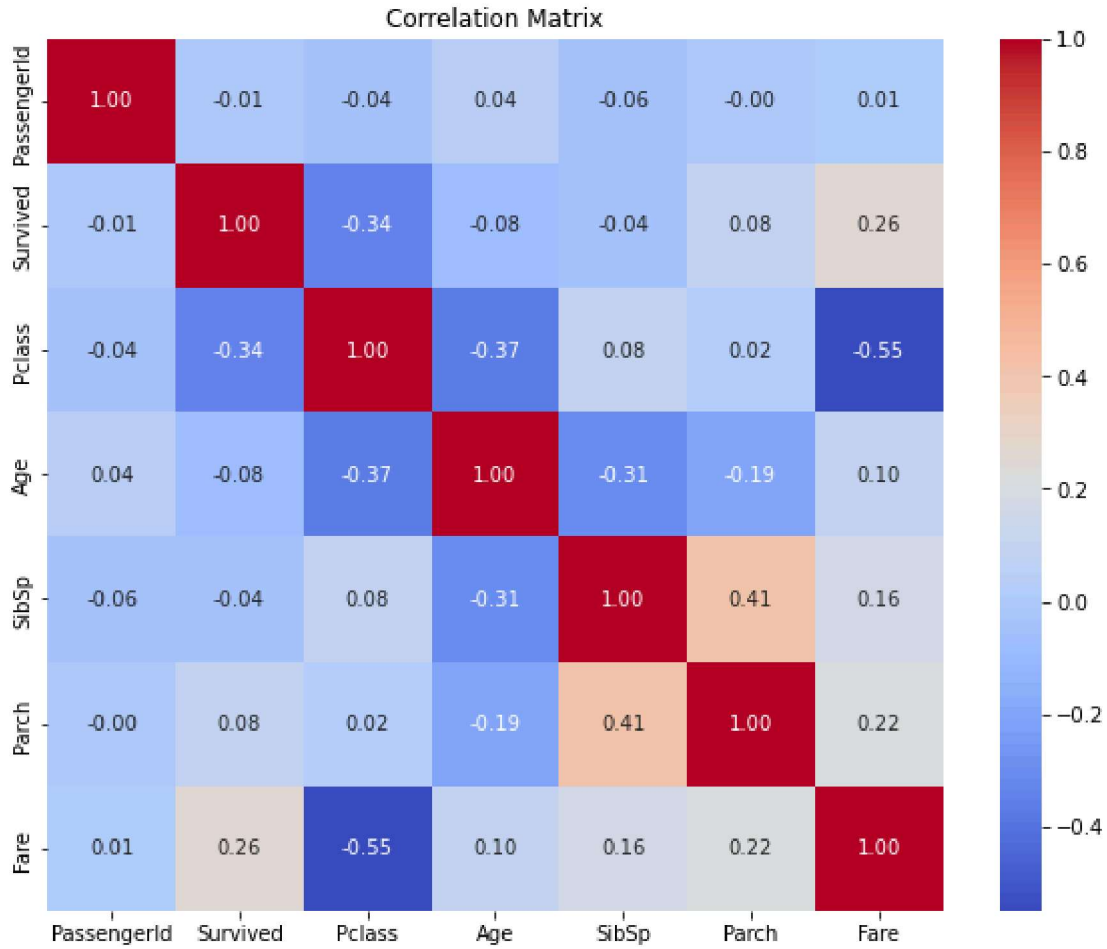
Survival Rate by Gender

In [10]:

```
plt.figure(figsize=(8, 5))
sns.barplot(x='Sex', y='Survived', data=titanic_data)
plt.title('Survival Rate by Gender')
plt.xticks(ticks=[0, 1], labels=['Male', 'Female'])
plt.ylabel('Survival Rate')
plt.show()
```



Correlation Matrix

In [11]:

```
plt.figure(figsize=(10, 8))
correlation_matrix = titanic_data.corr()
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Correlation Matrix')
plt.show()
```